

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 16: Loci (Locus and its Constructions)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Loci in a triangle

1. Statement 1: The point equidistant from three non-collinear points  $D$ ,  $E$  and  $F$  is the circumcentre of  $\triangle DEF$ . Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true **ICSE 2025, 1 mark**

### SOLUTION

A point equidistant from the three vertices lies on the perpendicular bisectors of all three sides. That point is the circumcentre, so statement 1 is true.

The angle bisectors meet at the point equidistant from the three sides, which is the incentre, so statement 2 is true.

**Answer: (a)**

2. The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes **ICSE 2024, 1 mark**

### SOLUTION

The circumcentre is the centre of the circle through the three vertices, so it is the same distance from each vertex.

**Answer: (b)**

## 2 Construction questions

3. Using ruler and compasses, construct  $\angle ABC = 90^\circ$  with  $AB = 6$  cm and  $BC = 8$  cm. Construct (a) the locus of points equidistant from  $B$  and  $C$  (b) the locus of points equidistant from  $A$  and  $B$  (c) the locus of points at the fixed distance  $OA$  from  $O$ , where  $O$  is the point common to (a) and (b) (d) the locus of points equidistant from  $BA$  and  $BC$ . **ICSE 2024, 4 marks**

### SOLUTION

Draw  $BC = 8$  cm, construct a right angle at  $B$  and cut off  $BA = 6$  cm.

- (a) Draw the perpendicular bisector of  $BC$ .  
 (b) Draw the perpendicular bisector of  $AB$ .  
 (c) The two bisectors meet at  $O$ . With centre  $O$  and radius  $OA$ , draw a circle. It passes through  $A$ ,  $B$  and  $C$ .  
 (d) Draw the bisector of  $\angle ABC$ .

Check:  $O$  is the mid-point of the hypotenuse  $AC$ , and  $AC = \sqrt{6^2 + 8^2} = 10$  cm, so the radius  $OA$  should measure 5 cm.

**Answer: the loci are two perpendicular bisectors, a circle of radius 5 cm and the bisector of  $\angle ABC$**

4. Using ruler and compasses, draw a circle of radius 4.5 cm and a chord  $AB = 6$  cm. (a) Construct the locus of points equidistant from  $A$  and  $B$ , and mark the point  $D$  where it meets the circle. (b) Join  $AD$ , construct the locus of points equidistant from  $AD$  and  $AB$ , and mark the point  $C$  where it meets the circle. (c) Join  $BC$  and  $CD$  and measure  $CD$ . **ICSE 2020, 4 marks**

### SOLUTION

- (a) The locus is the perpendicular bisector of  $AB$ . It passes through the centre and meets the circle at two points; take  $D$  as the point on the far side of the centre from  $AB$ .  
 (b) The locus is the bisector of  $\angle DAB$ . It meets the circle again at  $C$ , which is the mid-point of the arc  $BD$ .  
 (c) Join  $BC$  and  $CD$ . Because  $AC$  bisects  $\angle DAB$ , the chords  $BC$  and  $CD$  are equal.

**Answer:  $CD$  measures about 5.1 cm**

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